

Aim: EMBEDDED C PROGRAM TO ADD AN ARRAY OF 16-BIT NUMBERS AND STORE THE 32-BIT RESULT IN INTERNAL RAM

Output:

The screenshot displays the Keil uVision IDE interface during the simulation of an embedded C program. The main window shows the source code for 'exp13.c', which includes a loop to calculate the sum of an array of 16-bit numbers and outputs the result in three 8-bit segments to Parallel Ports 0, 1, and 2.

```
1 #include <reg51.h>
2
3 void main(void)
4 {
5     unsigned int i;
6     unsigned int array[5] = {0x1111, 0x2222, 0x8888, 0x4444, 0x0000};
7     unsigned long sum = 0;
8     for (i = 0; i < 5; i++)
9     {
10         sum = sum + array[i];
11     }
12     P0 = (unsigned char)(sum & 0xFF); // Lower 8 bits
13     P1 = (unsigned char)((sum >> 8) & 0xFF); // Next 8 bits
14     P2 = (unsigned char)((sum >> 16) & 0xFF); // Upper 8 bits
15     while(1); // Keep output stable
16 }
```

The 'Registers' window on the left shows the state of the microcontroller's registers. The 'Call Stack - Locals' window on the right shows the current function call and local variables. The 'Command' window at the bottom shows the simulation progress.

Register	Value
R0	0x00
R1	0x00
R2	0x00
R3	0x00
R4	0x00
R5	0x00
R6	0x00
R7	0x00
SP	0x00
PC	0x0009
STATUS	0x0000
SEC	0.00019450
PSW	0x00

Name	Location/Value	Type
MAIN	C:\0x0909	
i	0x0000	uint
array	0x0008	array[5] of uint
sum	0x00000000	ulong

Simulation progress: 11: 19085.54565800 sec L6 C:1 CAP: NUM SCRL OVR: R:WV